ActInf Livestream #048.0 ~ "Communication as Socially Extended Active Ir

Livestream | 1:06:16

FRIEDMAN: All right.

Hello. Welcome.

It is September 2nd, 2022,

and we're in ActInfLab

Livestream number 48.0.

Welcome to the Active

Inference Institute.

We're a participatory online institute

that is communicating, learning

and practicing applied active inference.

This is a recorded

and archived livestream,

so please provide feedback

so we can improve our work.

All backgrounds

and perspectives are welcome

and we'll be following video

etiquette for live.

Head to active inference dot org

or active inference institute

to come to our website.

Learn more.

We are in

active stream number 48 series.

We're going to be discussing the paper

communication

as socially extended, active

inference and ecological approach

to communicative behavior.

Paper in 2021

by Rémi Thyssen and Pierre Poirier.

The video is a total introduction

to some of these ideas.

It's not a review or

capturing every aspect for that.

You can read the paper and explore
some of the ideas.
We are going to talk about.
Aims and claims, abstract
roadmap, keywords
and bring up some notable points.
It is still
upcoming with the 48.1 and 48.2.
So people are welcome to
submit questions
or things to discuss,
write comments
and join the panels live if they want.
So
let's begin
with introductions and warmup.
I think we can share more
with our big question.
So Jean,
why don't you say hello
and introduce your big question?
TICKLES: Thanks, Daniel.
I'm Dina.
I'm Sharon, Calgary, and
well,
I could
I could read out the big question,
but it's pretty long winded.
So I'll just say that I'm always curious
about what happens
when active inference
isn't something that we look
at as a subject or a topic.
That is something that we actually
use or perceive as being around us.
And so I think the question
of whether or not
you focus on

what's inside of
an ecological space
or look at the ecology and the system
in which we believe active inference
exist this as and and
maybe within because we can the 2747
live stream
we kind of touched on
the idea could be both
if if we're going to
if we're going to expand our horizons
and look at
what kind of limits are
and describe that as an ecology.
My question then is
as communication and of communication,
what is the product of communication
and what is communication serve?
And so that's kind of like the questions
and that's
what in the next three life
means, we can start
maybe at least staking out
where that ecology begins and ends
FRIEDMAN: by.
Well, very interesting.
Thanks a lot.
Also for
so much of the preparation here,
we're still coming down off of 47.
And so I think it's going to
be interesting
to see how we approach
pragmatism in this discussions.
So the big
questions that
brought me to this paper
or that I was thinking about during

the paper were
what are the applications
and implications
of different interpretations
of communication?
Why does it matter
what would be the things in the world,
including the mind that would be changed
as a factor with or downstream
of how our interpretation
of communication is?
And then what are the causes
and consequences
of different perspectives?
We take on active inference,
where perspective also includes action,
propensity and engagement with.
So perspective is like
where one is situated and viewing.
And so it's the view from somewhere
from something
rather than a view from nowhere.
Slash everywhere.
And I think in that specificity
we can have a lot of discussions
about ecology
as a great system to learn from in yeah.
I think when we're outside it earth
we perceive ourselves
where we're we're framing
that would be the on this slide
that would be the first box.
That's our sense
that we're kind of outside of the system
looking in,
whereas the second box on the right
would be, well, we're embedded.
And now the, the, the,

the reason we're at
it is as the genie came out, it got out
and now it's all around us.
TICKLES: It's kind of like that.
The perfume is now being distributed
through the air
and now we're walking through it.
So which of those two things,
depending upon how we want to analyze
which of those two things
produce, what results sort of thing.
And I think we're both kind
of asking the same question.
If this then what?
If that, then what?
FRIEDMAN: All right.
I'll read the first half, the abstract
in this paper
we introduce in ecological accounts
communication,
according to which acts of communication
are active
inferences achieved
by affecting the behavior
of target organism
via the modification
of its field of affordances.
Constraining a target organisms
behavior constitutes
a mechanism
of socially extended active inference,
allowing organisms
to proactively regulate
their inner states through the behavior
of other organisms.
TICKLES: In this general conception
of communication,
the type of cooperative communication

characteristic of human
communicative interaction
is a way of constraining interaction
dynamics towards the goals of a given
joint action
by constructing
and altering
shared fields of affordances.
This account embraces a pragmatist view,
according to which communication
is a form of action,
aiming to influence
behavior of a target and stands
against the traditional
transmission view.
Sorry,
you just have to shut up here
because it kind of slid on my screen.
Where am I transmission to you?
According to which communication?
Fundamentally to convey information,
understanding acts of communication.
This act of inference
under this
ecological interpretation
allows us to link communicative
and ultimately linguistic behavior
to the biological imperative, minimize
free energy,
and emphasize
the action oriented nature
of communicative action
interactions.
FRIEDMAN: All right, nice.
Here's the road map,
and there's some light
sometimes consisting
highlighting at play.

The paper is prose based.
There are not any figures
or formal isms.
However many are suggested.
So we'll have an interesting time.
I think making some representations,
using words,
maybe some non linguistic communication
because there's so much to add.
It's not providing visual centers
as many papers outside
of especially philosophy do.
So first there's an introduction.
Then the central distinction or
dialectic or division is introduced.
The transmission
and the influencing
the and verses but or
the contrary is indications that we've
kind of explored this time.
That type of a dyadic
comparison is being brought
to two views of communication
to supporting witnesses join,
which are ecological interpretation,
the ecological developments and the turn
in that area and active inference.
And they're going to
then in communication, active inference
focusing on human
cooperative communication.
Describe how under ecological
and active inference starting points
justified starting points,
as they'll argue
that we can endorse a pragmatist view
at the
detriment to the transmission view.

Any other thoughts?
Yes.
TICKLES: Now, the authors are pretty clear.
They they
describe this paper
or these arguments as a sketch.
So when we say roadmap here,
I think that it's more of
it's not a
what a highly detailed, highly specified
zoom in on a on a you very precise
Google map.
This is kind of a back
of the napkin idea
for for a conversation starter.
And so I appreciate that
because I mean,
at least by the authors saying,
we want to get this down,
we want to stake this out
and then let people decide
how to come at this
empirically to maybe fill
in some more of the precision.
FRIEDMAN: Yeah, it's like planning a road trip
in its implications,
especially that
itself is an entire completed work.
Let's just jump to the aims.
Well, the top two aims
are some of the largest.
They summarize their work
as introducing
a pragmatist conception of communication
grounded in ecology of active inference.
So the three threads are pragmatism,
ecological interpretation
and active inference

about communication.
And they're going to argue
that all of communication be understood
as socially extended, active inference
be understood by whom, when,
and those are some of the points
it brings us to.
Secondly, there are some direct
contradictions slash tensions described
and contrasting
with the transmission view, which is the
the dancing partner of the advance
of the ecological pragmatic active view
is two
is by contrast
with the transmission view,
which can help a lot with literature
continuity
in in setting direction for research
field people
looking for transmission
view can see that there's a paper
from 2021
that's that's connecting
and finding concordance.
And as they mentioned
they will
stress in several different ways
that it's an initial account
with many gestured to next directions
that will get to any other thoughts.
TICKLES: Yes. Yeah.
I think without using these terms,
they're they're saying,
you know, there's
there are nodes out there,
there's interconnect,
there are potential connections

out there.
One way of doing
that is through a transmission line.
And what we would like to do is maybe
advance the idea that those links
are a little bit more embellished
than something simple.
And so,
again,
those are the terms that they use.
But with Daniel,
you're better at this than I am.
We have
a recent
live stream this year,
I think, where
we talked about offloading
as being a really important part
of extending the active inference.
You remember which one that was
kind of.
Yeah, yeah.
FRIEDMAN: Extent to which something was extended.
Let's look I f yeah.
It was with the cognitive offloading.
TICKLES: Cognitive all.
Communicate
this is 48 or extended
active inference constructing
predictive capacity
beyond Scholes number 41.
Right and that and that
and that live stream
there wasn't really
it wasn't anything in that idea that
the between this was like the whole idea
of how markup it may function
as a way of of of focusing

or bringing down the amount of noise
and raising up the amount of signal in a
in a space outside yourself.
There was no
we didn't have a lot of pushback
on how that theoretically would work.
So I don't think that they're stepping
into that twilight zone
by mentioning this.
FRIEDMAN: Nice. All right.
So they
make several claims
that are summarized here,
but we'll we'll deal them in
the sections in rough order.
Anything to add, though?
Okay.
TICKLES: These are just not really
going to come up yet.
FRIEDMAN: Yeah, these are some summaries
of some important claims,
but we're going to come up with them.
So. All right.
And then last,
last stop,
before we jump into the background,
which is one term
and then the paper itself,
what are the keywords in your eye?
TICKLES: Well, I mean,
we had dynamical systems model
in the last live stream, too.
So there's a lot of people that are
are depending on that theory
to advance
whatever arguments they want to make.
And the other
I think the other term

that comes up quite
often is joint actions
without really getting into some
hard specifics about what
that might look like on a
scale friendly level. So
I think when
the authors
join us,
I think both of those things are things
we'll have.
There's fertile ground there to
to grow some conversations.
FRIEDMAN: Awesome.
The one I would just want to
highlight is Dynamical Systems Model.
You you mentioned how
maybe just rephrasing
that people use it to
in a lot of different ways
and just that being raised
made me think about
what is being advanced by claiming that
alone or together,
something changes through time.
Does anyone disagree
and would it matter
that anyone disagrees?
But what does dynamical mean
when we're talking about time
dependent, real existing systems?
What does it mean that you have a system
or a model or a systems model?
What does that mean?
Because that's something
that's an artifact
or is it a, quote, mental model?
And so there's a lot to explore

about what
is being meant
by dynamical systems model.
We're thinking about time
or does it mean something more than that
continuous time, discrete time
implementation.
In fact,
there's so many aspects to it
and it's actually going
to come back at the very end.
All right.
And pragmatics,
also the top part,
you could mention anything otherwise.
Then how does this connect
to what we discussed in 47?
TICKLES: Well,
I kind of wrote this, so grab a read it.
Pragmatism in action,
especially the question
of what makes up an aim.
And I put in behind
that the act, the act, the process
and aiming,
which is the sense
aligning, which is
we all know as the generative model
given targeting with what there's
still a question of
what are we targeting with
when we,
when we put it all under one umbrella
called active inference,
a situational analysis tool.
This this is we've had lots
and lots of people
claim that that's what's

that's what active inference is.
It's a range finder
with a depth of filter
or is active inference
in a performance generator
something that enables as collection
given
to reframing or
depth stabilization or both?
Now, if you want to write
an academic paper,
it's way easier to just limit it to.
Active inference is a stabilize.
Now we have t01, two and so on,
but it's also
a way to
delimit that temporal,
spatial, temporal field.
So we just ignore that
because it's easier to write,
because if we do, that's
where the arrow of time,
meaning where
we're kind of more moving
with the flow and temporal depth
through the memorial
and across the spectrum, the spectral
range, whatever we arbitrarily pick
as our projection on integration space
must both be induced
or included
in whatever the representational is
if we analyze the entanglement
along with the attainment.
So what is the relationship is?
I mean, is the relationship
in the passing rings?
No, of course it isn't.

But those symbols still reinforce
what's going on.
So how can we just say
there's no representational ism in this?
It would it would be hard to do that
on realistic scans.
That would be pragmatics in a nutshell
to me.
FRIEDMAN: Very interesting closure.
TICKLES: Well, especially if you're
if you're recently involved in a
in a long lasting, long
term relationship. Right.
FRIEDMAN: It's so true.
Yeah.
The part I'll point out
is just the few
highlighted words on the top
with a lot of citations provided
that are going to be taking a pragmatic
federation approach,
broadly disparate interests and foci
that are being used by these authors
in their advancement
of an anti transmission view
distributed language
hypothesis, constraint view of language,
participatory sense making framework.
Maybe those fields are connected
in their citations or co-authorship,
but then the authors
are able to connect these
as kind of three people
pointing at the same process or thing.
And that's like a nice dense usage
that could be really catching people
up on a few ways
to navigate these topics,

because there's a lot of related topics.
It's like,
aren't we just talking about
how things are,
including how modeling is?
They begin
in the introduction
with an overview of the
layout of their paper
and they
describe, as they mentioned, the sketch.
It's a pragmatist account
of communicative behavior
grounded in ecological interpretation of
the active inference framework,
contrasted with the transmission
view of communication,
according to which the function
of communicative
behavior is to convey information.
So a claim is
that communication
is about information transmission.
Then they are contrasting that
in light
of ecological
psychology and theory broadly,
and also giving an active inference
twist slash theory, integration
slash introduction
of free energy
minimization or uncertainty reduction
as imperatives
or modeling imperatives
or existence imperatives.
Whereas some of these previous
threads, like ecological psychology
more neutrally,

wouldn't have
necessarily a formal
or a first principles way
to connect to modern applications,
though they may have a first principles
grounding and pragmatism.
So many of the
pieces have
already been assembled
in terms
of the ecological and pragmatic
and and all of these areas
coming together outside of the paper.
And so that will be a theme
what is linking out to other empirical
and philosophical literature.
What is active inference doing in this
mediation as a framework,
something that makes it
an organized
principle, organizing principle
in a way that merely
the literature corpus of these areas,
which in many respects
have been connected,
like the pragmatics of linguistics.
So what is active inference?
Doing anything to add on that?
TICKLES: Well,
again, I don't think it's hard
to argue that
to focus only on the string
and the two cups
is probably reducing to fork.
Or when we're talking
about communication, I mean, they're
they're entangled, right.
Like they're now tied together.

But what the authors
are trying to point out here
is that there also are signal
learners and signal receivers in it.
And as soon as you add
more layers to this of what's
actually going
on, suddenly an ecology emerges.
So let's step back
and look at everything that is possibly
participate painting in this,
not just the hardware.
And so again,
it will be hard to argue against that.
At the same time,
if you and I don't have all of the
all of
the things
that we need
to communicate from Davis,
California, to Calgary, Canada,
no matter how much we scream,
there is no communication.
There's no live stream 48 right now.
So again,
to dismiss a transmission view, I think
is also probably
not the best idea in the world,
because without that, what do you got?
FRIEDMAN: Yup, good point.
And they'll
bring up that sort
of enabling system perspective
in the conclusion very clearly.
Yeah, they set up the perspective
of communication as transmitting
information, symbolic information
or other kinds of information

influencing sculpting.
What is it influencing?
Is it modifying cognitive dials?
Is it modifying
the field of affordances?
Is it doing both?
Is it
any number of ways
that people have been studying
the nexus of communication,
intra species, interpersonally,
everything.
And you've done it a lot to say.
TICKLES: I'm going to say
if you read
if you've read anything
from Charles Sanders Peirce,
you would have to say it's both
like you.
I mean, if you're a follower,
you're not left with much choice
because there's a lot of in it's
nonsense going on
because it's a lot of interpreting and.
FRIEDMAN: Yeah, awesome.
All right.
Setting the context, do
you want to mention it or do
I mean to mention it? Go for it.
All right.
Well,
the first footnote is about
the notion of information.
And they describe different information
and uncertainty concepts
which we can address
in later discussions.
They do not dedicate a lot of time

and space
to fleshing out the transmission view
and different ways
that people have formally,
informally addressed it, causes
and consequences, etc.
However, they have provided
a lot of citations.
Then, in footnote four,
they are describing French
a writing in the 1890s,
which is a very interesting connection.
I don't know the details,
but some is provided in this section.
Footnote
It's an interesting path.
It would be fun to kind of go down more
because we've now brought back in Paris
to the discussion as others are.
And I think
walking the fine line
between
not using last names for equations
while respecting the process
and contributions of past
researchers and practitioners
in just different areas, will be really
part of what
active inference brings together.
All right, anything else.
I, I would say is that I think what this
maybe points out, again,
this is not what they wrote
in their notes, but
with every transmission,
there are things like
conflict embedded in that or complexity
embedded in that.

I mean, any
any good
writer knows
that you can't just transmit
a bunch of information.
There are
there are expectation issues around what
what are you moving towards
and why would I continue to communicate?
TICKLES: And so you have to have a context around
which that that continuing
regime of attention is held up.
FRIEDMAN: Yep, sure.
Attention regimes of attention
joint action.
Oh yeah.
So setting logic
here, they are going to flush out
two primary reasons to explore
theoretical alternatives to TV.
So not they're wrong.
They thought that.
But there are two reasons to explore.
We're at camp and we're going
to explore some theoretical
alternatives.
Firstly,
TV entails a commitment to an account
of the putative contents
conveyed via communication.
Providing such an account
is notoriously difficult,
especially if linguistic practices
cannot be invoked
as part of the explanation in
very deep
points
about what is linguistically accessible

or consciously
accessible, perceptible under different
situations, through different kinds
of memory, anticipation,
and all these features
which we know do play roles in
symbiosis.
And then the second reason
why to explore theoretical iterative is
how it's not clear
that communication conceived of
as the transmission of content
could be relevant
to the basic
allosteric processes
which drive the behavior of organisms.
So you might say bio semiotic li
the pancreas communicates
with different cells
in the body through hormone signaling,
but in a transmission to you,
would one be committed to the viewpoint?
Insulin means do X, Y or z?
To whom under what conditions?
In what ways is that a communication?
So especially at scales and systems
that aren't
quite literally humans
talking about language
which is quite common, then
there's going to be an account
that's non linguistic.
So to what extent are we
going to privilege or center
this type of linguistic discourse
and communication in a broader ecology
of communication?
TICKLES: So if the if the content by itself can't

or cannot
help us derive meaning,
then what
this essentially is saying is,
is that we need more than that
in order to get
the equals sign or symbol to
equal weight to meaning.
Right.
So that's all essentially what you're
trying to talk about here is
if we're going to go past TV y.
FRIEDMAN: Yeah, it could be a supplement to TV or
an extension of TV,
which, as I point out
later, can ever be denied or rejected.
So then pragmatism.
They trace a thread of scholarship
in which language is viewed as
a form of action.
And I'm reminded of
Professor Deborah
Gordon's paper
that can sustain an ant watching,
seeing the behavioral ecologist
as they're observing ant behavior,
the multiple stages and steps and ways
that knowing and acting and naming
of different kinds, names of activities
and then association of activities
as named with identities.
And that naming process
that comes in very late
in the game
in terms of the actions taken to prepare
the context for that observation.
So I thought that was a
very interesting link.

And here are those green
highlighted words from earlier.
So what else would you add about
pragmatism?
TICKLES: Well, I think what they were
moving towards here,
at least alerting us to,
is that the fact
that there are aspects
of a pragmatist view,
although it may sound
a little strange to say it at first,
I think is true,
at least with a pragmatist view.
You can now conjure things
that you can't do simply with content.
I mean,
I can have a whole bunch of ingredients
and not have a recipe,
but at least with a pragmatist view.
Now I can start experimenting,
and I think that's all they're
basically asking us to contemplate.
It's the experimentation piece.
It's not just
sort of leaving everything out on the
on the bench and going, Whoa, well,
I can take an account of this
or an inventory of it,
but not really do anything with it.
So that's
that's what I take away from
this part of the paper.
FRIEDMAN: Nice.
All right.
Here's a discussion around
what signals mean anything to add?
TICKLES: Not me.

It just read it.
It's pretty.
FRIEDMAN: Straight up. Yes.
Okay.
In this section
setting active inference
framework,
we've informally introduced
the ecological interpretation
of the active inference framework
in the terms of which we will formulate
a pragmatist conception of communication
in section
communication and active inference.
When is it of the active inference
framework, the genitive case?
When is it in the active
inference framework? The locative case
with the instrumental case?
There's so many
routes to explore with linguistic
and non linguistic communication.
Even only keeping with active inference
as the gripper and the drift right.
They provide a description
which is copied here,
and one key line
that they'll bring back
in service of communication
is organisms can reduce free energy
either by adjusting the sensory
input to the generative model,
quote, active inference, or by adjusting
the generative model to the sensory
input perceptual inference,
which correspond
respectively to action and perception.
So one could ask

whether active inference
only applies to adjusting sensory input
or what whether action
corresponds to active inference
and perception to perceptual inference.
That's one way to think about it.
It certainly is
one sense of active inference,
which is inference about action
planning,
decision making,
and also the consequences of action
being able to introduce preference
into future and current actions.
And we're talking about acting
inference also as a framework
for considering this view.
So
that was just the connection
with active inference there.
One connection a part is a part of this.
I ask you because I'll ask I may
ask the authors to do the dot one is
of this stuff
that they just wrote out here
because we've seen this many,
many times.
People just want to make sure
that they're following what
the three energy principle
up to this point has
has has generally been agreed to to mean
my question
from that
last organisms can reduce free energy
either by
so if I were to give you the metaphor
of getting the

toothpaste back into the tube,
how does communication serve
that purpose?
TICKLES: Right.
Because it's really,
really easy to squeeze it out.
But how does communication serve?
You get it back in
because
it it can.
But how does
how does what they're describing here
as an ecological view
speak to that?
FRIEDMAN: Let's explore.
TICKLES: Yeah.
They then go to
the ecological interpretation
of active inference.
So after introducing
just purely in principle,
using some active
inference, ontology terms and key ideas
like embodiment of generative models
and reduction of surprise,
or bounding that through variational
and expected free energy,
they continue to add in ingredients
to the recipe and
contrast a abstract theory
neutral description
of active inference, which is their
framing of vanilla or standalone
more somehow,
perhaps archetypal active inference
as being theory neutral
and hence compiled,
and therefore compatible

with many conceptions of cognition,
FRIEDMAN: pointing to a citation
where it's described as a state theory
compatible
with many different process theories,
which is why potentially someone
could, and probably has argued
for a transmission
view using active inference
as symbolic communication
or semantic communication.
There's probably so many
angles on that we could
bringing in the ecological papers,
citing some of the earlier
ecological psychology
and perceptual work
in predictive processing,
and then just connecting it
to more recent papers.
For example, Bromberg, Christine
Rietveld, Ramstad
many of these authors collaborations in
about the past six years,
anything else that.
TICKLES: Well,
I got some notes
that I want to
talk about on the top one,
but I think this is interesting because
it would be at this point where
if they wanted to introduce formalism,
which basically are
just a bunch of
symbols set up
operationally that they could,
but they haven't.
So how do we explain or

how do we take into account
all of those formulas that we've looked
at throughout a number of live streams?
If we're stop
if we're talking about something
that's ecological
because they are they're so
so are they floating around us
or are they only something that
we keep inside of a very contained,
very constrained, very conditions set
aspect of this of this diverse,
very complex
situation that we find ourselves under.
So yeah,
that when you as soon as you say,
oh, I would
I would assume
that one of the products of
an ecological interpretation
would be a hold my formalism
because you could derive this
pretty well.
I can't I'm
not Carl or or Thomas or one of those,
you know, super
math people, but they can.
So so what do we how do we
are we account for that.
FRIEDMAN: Yeah.
One note on
that is some of the famous models
of predator prey type dynamics are those
dynamical equations
are used in neuroimaging
to model, for example, winner
list competitions
among brain regions

or with different signaling regimes.
So in many ways
the SPM package specifically
as well as neuroscience more broadly,
many threads and precursors are here,
which is why I really appreciate that
by interpreting
the way you are
about how active inference
is being interpreted,
there aren't just
the bullet point or the visual abstract
of active inference.
If that is the active inference,
that is the interpretation
of the authors
using active inference.
They're working through it
and making a trace in a process
versus just saying, Well,
this is the standalone
author independent interpretation.
Well, I'm
not sure if that makes sense, but
all right,
this here's some text on a slide
to look at connecting
a spatial temporal structure
of affordances in ecological settings.
The concept known
as field of affordances
or the landscape of affordances
and different affordances,
different kinds of action capacity.
Introducing how salience and attention
and proclivity towards and affordances
is determined by its free energy,
by the free energy

it was expected to reduce
by the system
entity or by the modeler.
And there's a few more notes
on ecological psychology
and affordances, and I just.
Require that it's in the last bullet.
I think it's
I think it's really important
to again,
to go back to the
I want to ask you to go back
to a previous slide
that we don't carry around us formally
as some send those operations with us.
But humans become attuned
to social thoughts
by being immersed in
and participating
in shared social cultural practice
just very early in their development,
TICKLES: that that seems pretty obvious.
This immersion leads to the integration
of a social physics.
Okay,
well then
there is math involved in this,
which is the set of structured
and systemic
social contingencies available
in an environment of development.
So even if we can't identify
what those formal isms are,
those formal isms must be present
in order for this early
this thing that we call early
development and integration to exist.
It can't just be met.

It can't just
we can't just say,
Oh, it's there, but we can't explain it.
There actually
is something
symbolic and representational
need that
even if we can't conjure it ourselves.
There are
there are those two aspects
which we need to to include.
Right.
Just because I don't
I don't think of it as map doesn't mean
the map isn't present.
So I think that I think that's
I think that's
a really important one here
in terms of making their argument,
even though it's only collages,
there's still a lot of formalism
that that would seem to be abstract
but presents as as something real.
So,
yeah,
my little $0.02.
FRIEDMAN: Yes, that's
interesting direction
and made me think about attractors
dynamics gravity in social settings
and that puts quite a
spin on Bayesian mechanics
of social physics.
And social physics
has been approached from
the classical billiards case,
from the statistical
thermodynamic informational case

and probably
more speculatively
by some from the quantum case.
So it's those three mechanics, classical
statistical thermo info and quantum.
TICKLES: Yeah. Three, five. How many there are
are what are
being proposed in their integration
by Bayesian mechanics,
which we'll be discussing in
the next livestream actually in 49
with some really interesting formalism
to reenter into.
So I think it's like seeing
what's not there but seeing depth
in what they're pointing to, even
just by
discussing integration
of social physics.
FRIEDMAN: We don't need to provide 100 citations
to social crises or tragedies.
It is just
what is gestured towards with what
the domain of the work is.
All right.
In section communication
as socially extended active inference.
Here's some quotes.
Anything you want to add?
TICKLES: I just want to read bullet
through communication
is that's understood as we might
what we might call socially extended
active inference situations
where the organism
reduces free energy
by influencing the behavior
of another organism in such a way

that its sensory input
will correspond to the projections
of its generative model, thereby
regulating its own internal states
by the behavior of the other organism
other than through formalism.
How could we prove that
would be my question.
FRIEDMAN: Yeah, great question.
And
what if well,
what if there are many routes
and one can ask somebody to stop
writing on the sidewalk with chalk
or they could just wash it
while they're gardening,
or they could go in so many ways.
It's like the adjacent
possible of the social physics
is open in a way that
contrived or designed settings art,
which isn't to say
only useful or educational,
but that's not the same. Right?
TICKLES: The only way you could prove it, though,
I think, is to represent
notational symbolic
operations.
Like I don't know how,
because every time you try to repeat it,
someone like Dean comes along and says,
Oh, I think I know
what the game is here.
I'm going to disrupt the game right?
So that there really is only one way
to prove it.
And so, yeah,
I like that they put this in here now

and one of our colleague, Stephen
Dikes, talks
about the prefixes
added to active inference.
And so, again, do we need those prefixes
in order to be able
to explain our argument?
Is it
for the convenience of our argument?
Where did those things really happen?
And can we say
can we say they really happen
because we can prove it?
FRIEDMAN: All right, nice.
The following section
is some cases of communication
and three areas are highlighted,
some cases of communication.
And for those cases, one could ask
what is or could be
a transmission view on that setting?
Or How have people framed a certain case
or instance
or type of communication
as transmission view?
And then how is an ecological active
inference interpretation
changing that challenging?
Not going beyond it.
And three areas that they are looking at
are aggressive behavior
mating or sexual reproductive
signaling and alarm signaling.
And they write about
in each of these cases how one might see
the type of behavior exhibited
there as kind of
a canonical signaling case.

Like when you look up signaling,
you'll find these keywords describing it
as categories
of signaling and communication.
And one very excellent
book to follow up
on is how in long of those 2012 book
studying human behavior,
how scientists investigate aggression
and sexuality.
There are two of the go
to areas in behavioral study.
Two of the
go to approaches
and phenomena and Framings of
Action interaction.
And this is
a very
multi and transdisciplinary book
that addresses
in the context of some recent
historical debates
around the relevance
of genetic inheritance.
Ecological inheritance,
social structuring,
all these different topics
approached in a really edifying way.
I think it speaks to the gripper
and the being gripped,
which is kind of the role
of what history
and philosophy of science
has been doing
in a disciplinary academic way
with a certain approach
and time frame and feedback method.
GO okay.

Human cooperative communication,
anything that.
TICKLES: I'm always curious about
and I will put a little note
on on the actual page,
my copy of the paper
about cooperation
and I'm no expert on cooperation,
so I'm sketching this out
like the authors are sketching
certain things out, but
I think
this is what I wrote.
Let me see here.
I got
neither the
transmission version
nor the ecological feel
of affordances model
and say for certain
how long the cooperative state
will exist for
which requires
a different type of analysis
of the one that confirms
that communication lines are functioning
regardless of function type.
So stepping back and saying
that there is a type of communication
which we'll call cooperative is great,
we can
we can say that there's a type here,
but just naming
it doesn't
mean we've actually tamed
or been able to explain why
that type exists.
So that's one of those things.

Again, when we have the authors
with this
like to chat about that
a little bit more.
FRIEDMAN: Yeah, what that makes me think about
is even non
communication
can be considered part of the
perhaps shaping the
landscape of affordances,
perhaps being communication
of non linguistic type.
And so how could we have
a model of communication
where the special cases
of which there are infinite
special cases
are communications
in their embodied sense.
And we can also abstract
to people who only communicate
through social networks,
through breathing the same air,
all these kinds of very loose,
only ecologically aligned,
if only on earth aligned
views of communication.
Seeing that as like the fields
from which different forms
of communication
of cognitive entities arises from,
rather than something
symbolic and transmits, of which
could only be instantiated
kind of from a top down
by a symbolic subsystem
or sub functionality
which isn't denied by a bottom up view.

But it's contextualized
as part of a process that
includes the ecology
and the shaping
of the field
of affordances
and landscapes,
and the stigma of g
as something fundamental.
Now, much more
TICKLES: the less with the ecology,
as the context does, act
as a constraint.
So yeah. Okay.
FRIEDMAN: All right. Conclusion.
So they introduced
a pragmatist conception of communication
grounded in
an ecological interpretation
of the active inference framework.
There's a lot more to develop.
So where do papers end?
Where do projects begin?
Right.
And then they mention
two different directions,
which are relatively well confirmed
empirical results.
First is the work on joint attention
and seeing joint attention
as a relevant aspect
in relationship
to coordination on landscapes
and relevant for language development.
And then the second point
of empirical evidence
that they want to bring in
is neuroimaging data

showing co activation or partnership
in some way of different parts
of the body
and brain
associated with certain actions
that seem to be linked
in a way
that's best understood
or of course, screened
as being ecological or embedded.
And then here is the
part I mentioned earlier to conclude,
it is trivially true
that pretty much
any episode of
communication can be described
as a transmission of information,
as suggested in the section.
Some cases of communication.
So no bird call, no bird call.
Hard to get around that one.
In practice,
maybe there are
some theoretical tricks, but
what if communications pre-planning
really hear a bird call in 5 minutes?
Really want that to happen
or make that happen,
and then what happens?
However, there'd be no communication
if there was no information.
In some sense
that is some form of association
between a signal and an inference
which constitute
general ecological information.
Some ideas this person triad
objective discussion we had in 47.

So this is the bio semiotic ecological
angle that we
came from last time
and this paper
is going to really highlight
the ecological and the free energy
minimizing aspect via
a field of affordances, modifying
through stigmatic modifications
or other types of communication.
Maybe just making your body visible
in a certain way
and using
that as part of a joint,
socially connected
free energy minimization,
pointing the way towards models
that would do that.
But not as the only end point,
but it's in a framing
where models could conceivably do that,
like through software packages and model
design processes.
TICKLES: I sometimes choose words,
not carefully,
and I know that I don't think
that the word trivially
was placed in that last bullet as a as a
as a way of provoking.
But maybe it was I think basically
it's basically
true, but saying
it's trivially true is like saying,
okay, well,
a math proof is tricky
because it just is there.
I think it's sometimes
to to

overstate
how some of the basics matter.
That doesn't mean
get locked in just and start
because you got the basics right.
I mean, there's a whole bunch of stuff
that pops out of that
and they include that too.
If you're looking at something
realistically, however, I don't know.
I don't know
if it's very trivial or not.
Like I don't know that it's that common.
Some of these more interesting
mathematical proofs
we could say they are
because they've already been proven,
but that doesn't mean
everybody has access to it
or even understands
the importance of it.
So that's just a little side comment
because I know
like I catch myself regretting sometimes
describing things in certain ways.
All right.
FRIEDMAN: To defend the trivial,
I looked up the definition
I really wanted to know.
It turns out it is originating
from Trivium,
which is
an introductory curriculum
at a medieval university
involving the study of
grammar, rhetoric and logic.
So maybe it's it's so trivial.
It's just the kind of trivial details

that you end up spending
quite a long time working on.
TICKLES: Okay, that's good defense.
FRIEDMAN: That would be
that would be a funny interpretation.
You know, bring back the Trivium.
TICKLES: Where.
FRIEDMAN: Our hope
is that the model describes inspires
experimental designs
that can lead to such theoretical
or applied treatments. Okay.
What are your thoughts?
TICKLES: Well, as I wrote here,
I think that means keeping active
inference on the bench and constraint
to a control setting.
I mean, again, if you want to be able
to compare to a control,
this is this is your next step.
However,
since when does setting a condition
deliberate communication to do
a communication does
which is released
understanding and not just aggregate it
for the sake of control.
And I wrote a time to break up
meme watchers.
Then we've got more
direct experience with,
with writing about active inference
and needs than I. Yes.
FRIEDMAN: So that's the slide.
TICKLES: You can go.
FRIEDMAN: Yeah.
Well just wanted to kind of explore
this and

leave some open space to go into it
more. Yeah.
TICKLES: Building on the work of our Mridula,
who is a professor in Monterey,
California.
Yeah,
she had written about memes
as quasi arguments
and about the kind of space
that opens in
between in
enduring the meme
and specifically image
meme, interpretation
or interpolation process.
So in addition to
or an alternative
to the purely dynamical, descriptive
and evolutionary fitness oriented
meme definition commonly used by Dawkins
and all since this is a very media
and cognitive interpretation
of artifacts and through a paper
in 2021 and in 2022,
we used some previous work
that had been done
by Equity Hall at all 2020
on a three tier model of
how to frame complex ecosystems.
FRIEDMAN: The kinds that people
are usually talking about
when they mention ecosystems
like the non metaphor.
If there were a non metaphor
and it wasn't just app ecosystems
and social ecosystems,
the one with, you know, the trees, but
don't they all

three layers
being an instrumental layer
of pure observations,
basically mapping on to observe states,
then a contextual layer
which can bring in some knowledge
about how different measurements
or inferences about
measurements are linked,
like two common entities
and then using this type
of statistically inferred
information
and augmented information
from the contextual layer
to do inference on pure, on observables,
those might be things
that are observable,
but you didn't observe
like the temperature at some point
where there wasn't a thermometer
or the true on observables
could be like propensity of this mile
square of land
to be used for this purpose
in the next five years.
But that's not something
that could be measured.
There's no measuring device
other than the modeling process,
but it's an important modeling process.
And they did
amazing work
connecting complex systems and ecologies
and on the rise in term
and in spirit
is discussions of digital ecosystems
and ecological approaches

to communication,
as noted by the authors here,
and hence by extension
ecological approaches
to digital communication.
And so we were able to modify
or to translate and respect
the ecosystem
integrity three tier model
provided by those authors onto,
the kinds of digital
discourse analysis platforms
that would be aligned
in that
ecological interpretation and way.
So I think there'll be a lot to explore
maybe now, maybe next time.
So go ahead. Yes.
TICKLES: Quick.
Just quickly,
so would you describe what you did
from a
in transition to be as an in action?
FRIEDMAN: It was in an action in the slides
software of one type.
And then it becomes
a different in action
when it is shared
as an asynchronous file
or through live communication
with a colleague.
It's a different in action
when the preprint is open
source and available immediately
upon completion.
There are many levels of and
types of in action.
All right,

then it closes
with a discourse from Squirms Screams.
2010 suggests
that in any complex organization,
different highlighting
information transmission and processing
and coordination of action
may not be entirely separate
gasp
screams 1996 and storms 2010.
These are actually really
excellent books.
I hope that is not taken as dismissive,
but it goes to show what views
are seen as and justified
to be often needing support
and what that means or why that behavior
is that way.
What is that linguistic
strength for in this
quote from 2022 written
originally in 2010?
And then the authors
write What fundamentally matters
is the control of behavior
and not the transmission of information.
Even if the control of behavior
may be described as a form
of information transmission.
So very much a syntax semantics question
How many bits is the file?
What does the file mean?
Syntax and semantics in the specific
unfolding of the interpretation,
you can take a top down overview
like 55% of people who did
this also do that.
And then there's a bottom up

cognitive view,
which is the ecological
embodied engagement with the image mean
or with something like that,
with an animal signal or mark or smell,
and then
to paraphrase
Kearns, the information
transmission gloss,
footnote 35 with some Egan citations,
the information
transmission gloss
on communication
and communicative behavior,
if it is to prove useful at all,
might be thought of as an aspect
of the control of behavior
rather than the other way around.
Reversing primacy but not denying
we propose
they can see if in communication
in such a way
provides better insights
into the pragmatic nature
of communicative behavior
and its evolution
as a free energy, minimizing activity
and a favor.
What do you want to add?
That's basically it.
We have a few things
written down to 48.1, but
what would you like to add on that?
TICKLES: I mean, what I like about the conclusion
is that they say I'm
I'm taking and paraphrasing in my own
thinking, in my own
words, just because you transmit doesn't

mean that you're actually
sussing out or determining or
or confirming the difference
between play as one kind of action
playing and then played.
So they really want to incorporate
that temporal piece,
which I think is really smart,
which again,
I'm a
I'm a big believer
that there is a logic that
supports that, which limits
which direction you're necessarily
asking to look at time.
Are you going with time
and only one direction?
Are you enacting it
or are you delimiting and being able to
place some sort of
gap between the memorial
and the projection?
And the other thing is,
I think
when we walk
when we walk into these situations
and see them as ecologies,
we are going to miss things.
So I think we just have to accept that
there will be laterals and collaterals
on the margins that will will sort of
bleed into what
our regime of attention is.
And so I think
if you only focus on the transmission
my turn, your turn, my turn, your turn.
You don't have a hope of trying
to bring those very important things

that that are essentially building up
whatever you are
taking away from this
communication exercise.
So, again,
I think that's something important
that they raised in sketching this out.
There's actually a benefit or a feature
in not filling it in to it
with too much detail,
because how could they
every time
there's a communication act
going on or a dyad
happening, the openness of that
means there's
going to be some scale friendly follow.
It can't be predicted with certainty.
So that
I really like their conclusion actually.
So
yeah, there's my,
there's my thoughts
in those last couple slides.
FRIEDMAN: A lot to think about,
I guess, and discuss
in the coming weeks.
I think a few
last pieces here.
Seeing transmission as enablement
for semantics
and connecting that to the person triad
and objective logic
and as a escaping the nosedive
of description, ism and reductionism
by holding space for the pragmatics
if nothing else
of higher order

levels of analysis and course screening.
So whether you need to appeal
to computational resources,
we won't be able to finish the analysis
or other kinds of pragmatics,
including meaning as pragmatics
in the socially extended
cognitive space.
Then there's a very rich view in which
we don't need to like reject
digital signal processing
in order
to have an ecological interpretation.
I think there's a a few other
areas we can go
into.
So with that
we have
there's other relevant works to share.
Okay.
Any final, final thoughts?
TICKLES: No.
I mean, what I appreciate is how
how smoothly and quickly we're able
to kind of go through this paper.
I think that's a testament to
the authors
being clear about what
they wanted to ascribe
and what they
wanted people
to be able to extend off of.
So it kind of reinforces
their idea
of the extended active inference.
Instead of them
having all the answers,
they just

planted some seeds
and let people kind of take it
off in different directions,
which I think
maybe was one of their goals.
FRIEDMAN: That's very nice.
Totally agree.
Agreed that it's very readable.
Clear paper
that laid out a nice path with berries
to pick
from a lot of footnotes
and paths to explore and.
It is going to be great to discuss it
in the coming weeks.
So Dean,
thanks so much for the preparation
and for joining live
and we'll see you very.
TICKLES: Same back right back at you, buddy.
Take care.
FRIEDMAN: Peace.